

Topic 6 – Groups in the periodic table

Group 1

Students should:	Maths skills
6.1 Explain why some elements can be classified as alkali metals (group 1), halogens (group 7) or noble gases (group 0), based on their position in the periodic table	
6.2 Recall that alkali metals: a are soft b have relatively low melting points	
6.3 Describe the reactions of lithium, sodium and potassium with water	
6.4 Describe the pattern in reactivity of the alkali metals, lithium, sodium and potassium, with water; and use this pattern to predict the reactivity of other alkali metals	
6.5 Explain this pattern in reactivity in terms of electronic configurations	

Group 7

Students should:	Maths skills
6.6 Recall the colours and physical states of chlorine, bromine and iodine at room temperature	
6.7 Describe the pattern in the physical properties of the halogens, chlorine, bromine and iodine, and use this pattern to predict the physical properties of other halogens	1d 2c
6.8 Describe the chemical test for chlorine	
6.9 Describe the reactions of the halogens, chlorine, bromine and iodine, with metals to form metal halides, and use this pattern to predict the reactions of other halogens	
6.10 Recall that the halogens, chlorine, bromine and iodine, form hydrogen halides which dissolve in water to form acidic solutions, and use this pattern to predict the reactions of other halogens	
6.11 Describe the relative reactivity of the halogens chlorine, bromine and iodine, as shown by their displacement reactions with halide ions in aqueous solution, and use this pattern to predict the reactions of astatine	
6.12 Explain why these displacement reactions are redox reactions in terms of gain and loss of electrons, identifying which of the substances are oxidised and which are reduced	

Students should:	Maths skills
6.13 Explain the relative reactivity of the halogens in terms of electronic configurations	

Group 0

Students should:	Maths skills
6.14 Explain why the noble gases are chemically inert, compared with the other elements, in terms of their electronic configurations	
6.15 Explain how the uses of noble gases depend on their inertness, low density and/or non-flammability	
6.16 Describe the pattern in the physical properties of some noble gases and use this pattern to predict the physical properties of other noble gases	1d 2c

Suggested practicals

- Investigate displacement reactions of halogens reacting with halide ions in solution.